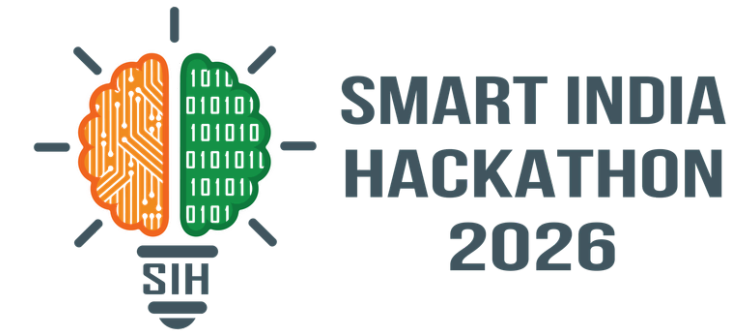


SMART INDIA HACKATHON 2026



- Problem Statement ID – **SIH26127**
- Problem Statement Title- **City-Wide AI Engine for Multi-Camera ANPR Trajectory Tracking and Urban Traffic Analytics**
- Theme- **Smart Automation**
- PS Category- **Software**
- Team ID-
- Team Name- **SylRak**



Proposed Solution: City-Wide ANPR Trajectory Tracking & Traffic Analytics Engine



>90% Multi-Frame OCR Fusion

Accurate plate recognition in real - world conditions



Cross Camera Identity Matching

Link sightings by time and location



Per Camera Confidence Scoring

Reliable and Explainable results



Real - Time Traffic Analytics

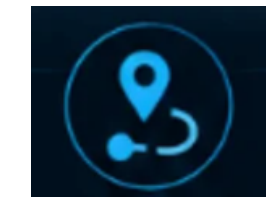
Density, congestion and anomaly alerts



1. CITY-WIDE FEEDS
Ingest ANPR/CCTV feeds from all cameras across the city



2. AI PROCESSING
Multi-frame OCR fusion, cross camera matching and confidence scoring.



3. TRAJECTORY TRACKING
Link plate sightings by time and location to reconstruct vehicle routes



4. TRAFFIC ANALYTICS
Live density, congestion and route - anomaly alerts on a GIS map

BRIEF INTRODUCTION TO OUR SOLUTION

The **SylRak City-Wide AI Engine** provides multi-camera ANPR trajectory tracking and urban traffic analytics. It integrates seamlessly with **existing municipal CCTV & ANPR networks** to track vehicle movements without requiring expensive new hardware infrastructure.

TRACKING & ANPR PIPELINE

High-accuracy **YOLOv8 & Deep OCR** extract license plates, vehicle color, and make. Spatio-temporal filtering performs **cross-camera re-identification** to reconstruct vehicle trajectories across non-overlapping views on interactive GIS maps.

Machine learning models analyze citywide flow, measure **Origin-Destination (OD) density**, and flag suspicious route anomalies in real time.

CENTRALIZATION & ANALYTICS

A centralized **GIS Command Dashboard** computes real-time traffic heatmaps, optimizes signal timing, and provides law enforcement with **automated watchlist alert notifications**.

TECHNOLOGICAL STACK



Python



YOLOv8



OpenCV



FastAPI



PostgreSQL



React



Kafka



Docker

1 Multi-Camera Data Collection

CCTV and ANPR cameras across the city capture live video feeds and detect vehicles with license plates.



Camera 1
(Sector A)



Camera 2
(Sector B)



Camera 3
(Sector C)

2 High-Accuracy ANPR & OCR Engine

Deep learning model detects license plates and performs OCR with >90% accuracy, handling real-world challenges (lighting, weather, blur, angle, dirt, damage).



Plate Detection



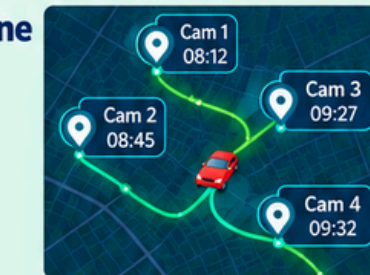
OCR Model

KA 01 AB 1234

Extracted Plate Number

3 Trajectory Reconstruction Engine

Links the same vehicle plate across multiple cameras over time, reconstructs the complete travel path and shows the route on a GIS map with timestamps, direction and camera locations.



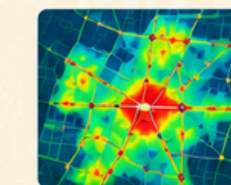
Vehicle History

- Cam 1 – 08:12 Sector A
- Cam 2 – 08:45 Sector B
- Cam 3 – 09:27 Sector C
- Cam 4 – 10:32 Sector D

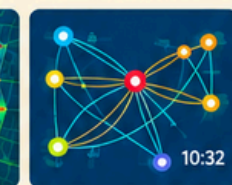
4 Macro Traffic Flow & Movement Analytics

Analyzes aggregated camera data to compute city-wide traffic dynamics:

- Traffic density measurement
- Origin-Destination patterns
- Congestion bottlenecks
- Real-time heatmaps & flow trends



Traffic Heatmap



OD Patterns



Congestion Detection

5 Alert System

Flags blacklisted vehicles and suspicious route anomalies in real time.



Blacklisted Vehicle Alert
Match found with watchlist / blacklist



Route Anomaly Alert
Unusual route / deviation detected

6 City Traffic Analytics Dashboard

Centralized, GIS-integrated web platform for real-time monitoring, insights and better decision-making.



Live Map & Heatmaps

- Average Vehicle Speeds
- Route Densities
- Traffic Flow Trends

TECHNICAL FEASIBILITY

-  Proven technologies: YOLOv8 for vehicle/plate detection and PaddleOCR for plate recognition
-  Existing Indian municipal ITS/ATCS control rooms provide live RTSP camera feeds
-  Compatible with diverse camera models and real-world conditions
-  Edge + cloud architecture enables real-time processing with reduced bandwidth



Cost Viability

Leverages existing camera infrastructure — no major new hardware required.



Deployment Feasibility

Can be piloted at a few junctions and scaled city-wide in phases.



Real-World Impact

Provides actionable insights for smarter, safer and more efficient cities.



SCALABILITY AND VIABILITY



Camera-agnostic: easy integration of new CCTV/ANPR feeds



Modular pipeline: detection → OCR → matching → trajectory → analytics



Scalable from single junction → zone → city-wide deployment

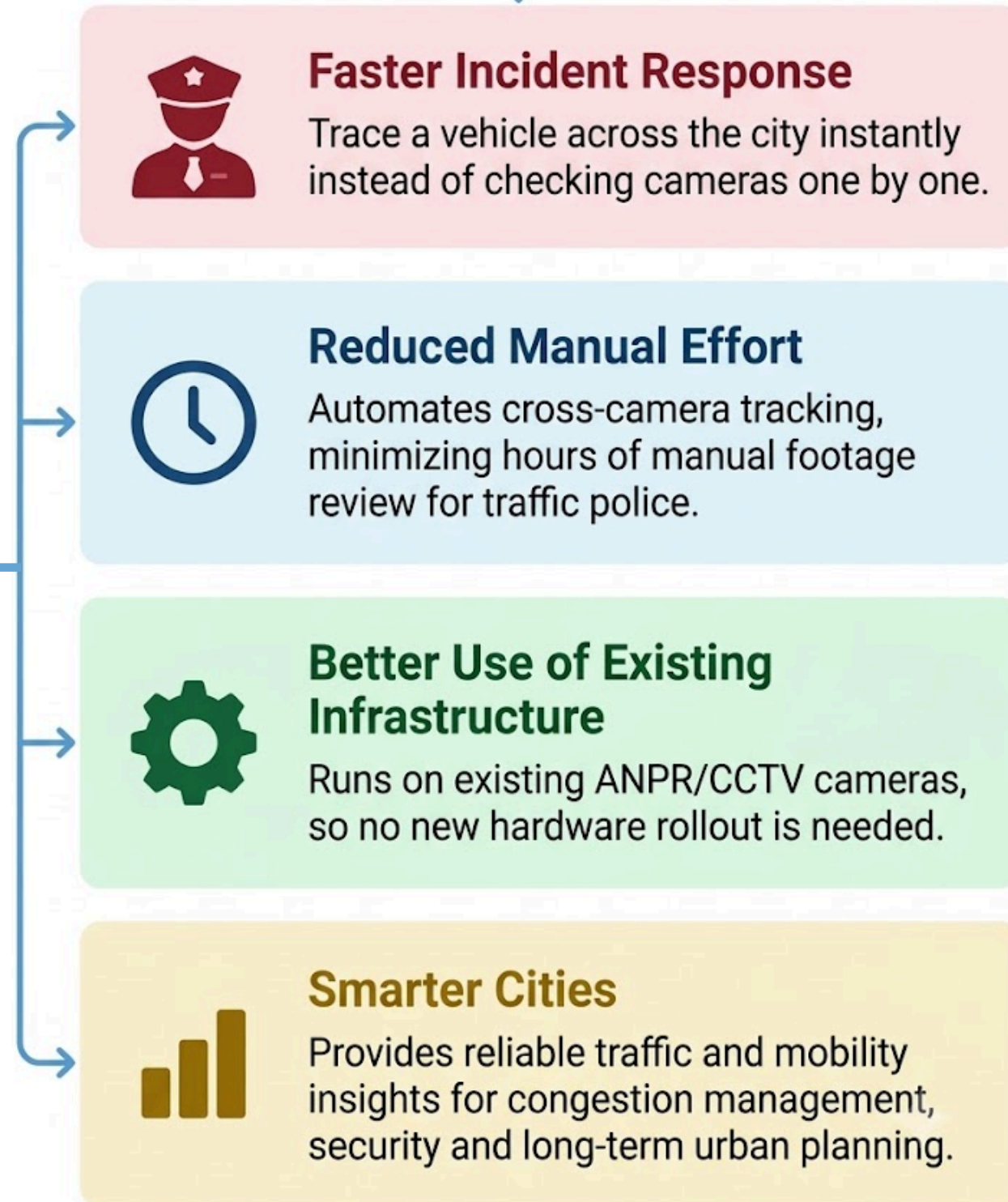


Uses existing infrastructure, reducing deployment cost

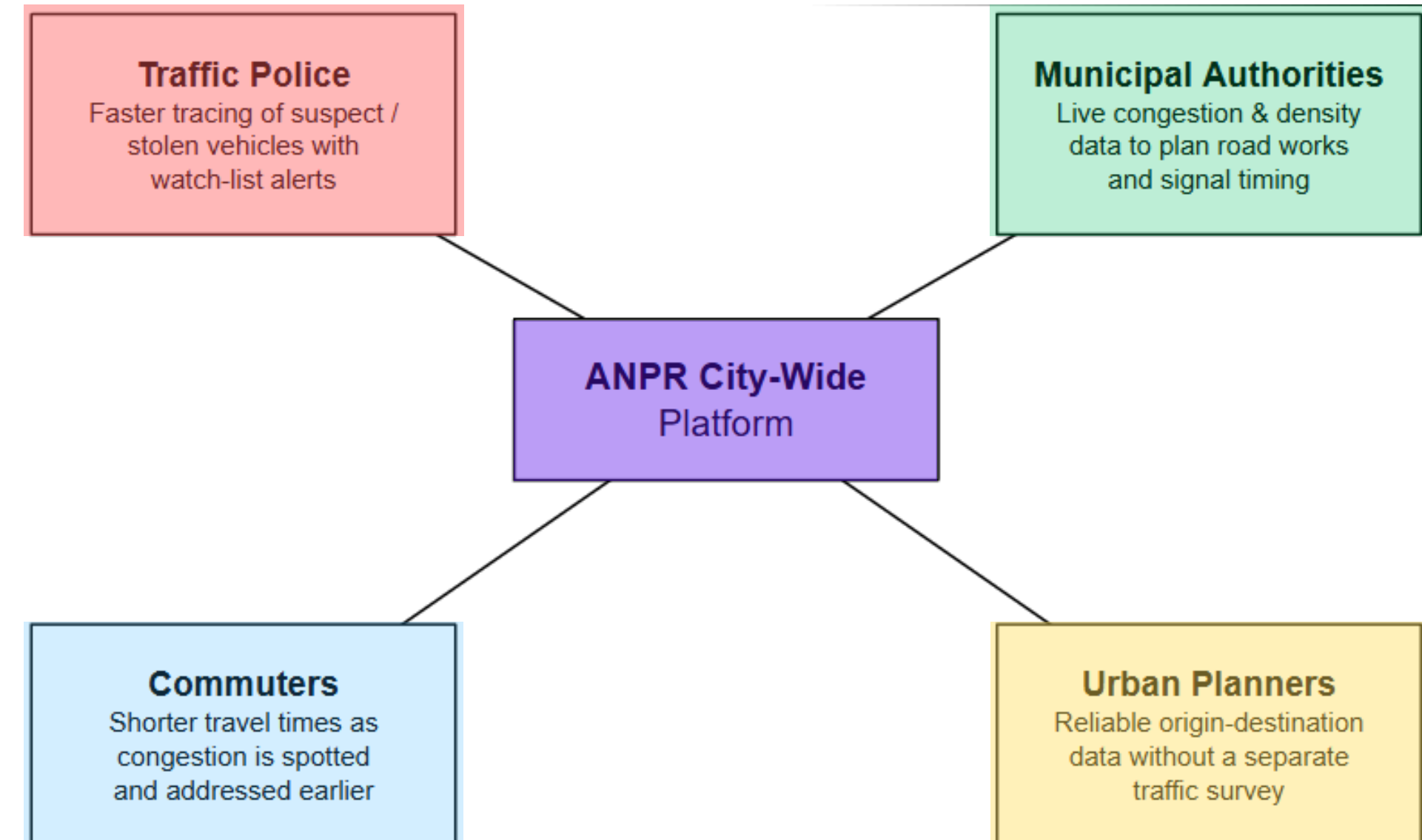


Supports multiple use cases: traffic management, congestion monitoring, route-anomaly detection, law enforcement and urban planning

IMPACT



BENEFITS



- Automated Vehicle Access Control using License Plate Recognition [Link](#)
- Official Ultralytics YOLOv8 Documentation [Link](#)
- TrOCR Research Paper [Link](#)
- Cross-Camera Vehicle Trajectory Reconstruction — 2026 [Link](#)
- Automatic number plate recognition (ANPR) in smart cities: A systematic review on technological advancements and application cases [Link](#)